

**Time:** 5 minutes**Outcome:** Analyze Collusion, Cartels & Prisoner's-Dilemma Incentives.**Difficulty:**

Intermediate

Collusion, Cartels & Prisoner's-Dilemma Incentives



THE CORE IDEA

A cartel coordinates output restrictions and higher prices. A prisoner's dilemma arises when each firm benefits from undercutting regardless of its rival's action, yet mutual undercutting leaves both worse off than cooperation.



HOW TO RECOGNIZE IT

- Compare a firm's payoffs separately for each possible rival action.
- Undercutting is dominant if it pays more in both comparisons.
- Check whether mutual undercutting pays less than mutual cooperation.



WATCH OUT

A jointly profitable agreement may be unstable when each firm can gain by breaking it. Compare individual incentives, not just the sum of payoffs.



WORKED EXAMPLE: INDIVIDUAL INCENTIVES VERSUS JOINT GAINS

Payoffs: (Row firm, Column firm)

	Keep agreement	Undercut
Keep agreement	(6, 6)	(2, 8)
Undercut	(8, 2)	(3, 3)

If Column keeps the agreement, Row gets 8 by undercutting rather than 6 by keeping it. If Column undercuts, Row gets 3 rather than 2 by undercutting. Column faces the same incentives. Both undercut and receive 3 each, although keeping the agreement would give 6 each. Individual incentives can undermine a mutually beneficial agreement.



CHECK YOURSELF

If the other firm keeps the agreement, why is a one-time promise to keep it vulnerable to undercutting?

**READY?**

Return to the game and master this concept.